Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

ACH101S – ALLIED CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Metal ion present in phorphyrin ring of chlorophyll is
   a) Cu b) Mg c) Fe d) Zn

2. NH_3 is
   a) Neutral ligand b) Positive ligand
   c) Anionic ligand d) Can act as both (a) & (b)

3. d & l – tartaric acids are
   a) Enantiomers b) Diasteriomers
   c) Geometrical isomers d) Both a & c

4. The hybridisation of carbon in methane
   a) sp^2 b)sp^3 c) sp d) dsp^2

5. Which among the following is a weak electrolyte
   a) Acetic acid b) Sulphuric acid
   c) Hydrochloric acid d) Nitric acid
6. the order of ester hydrolysis is
   a) 3 b) 2 c) 1 d) 0

7. The cathode used in the polarography
   a) SCE b) Glass electrode
   c) Platinum electrode d) DME

8. The reference electrode used polarograpy is
   a) SCE b) Glass electrode
   c) Platinum electrode d) DME

9. The number of sulphur atoms present in sulphafurazole is
   a) 4 b) 3 c) 2 d) 1

10. The number of sulphur atoms present in prontosil is
    a) 4 b) 3 c) 2 d) 1

II. Fill up the blanks

11. The ring system present in hemoglobin is_____.

12. Meso tartaric acid does not show _____.

13. Efficiency of heat engine is always _____.

14. In polarography, _____ is used as an indicator electrode.

15. Vat dye must _____in water.

III. Match the following

16. O_2 - (4n+2)

17. Aromatic - Constant heat

18. Adiabatic - Residual current

19. Polarography - Prontosil

20. Sulphadrug - Diamagnetic

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Name the following complexes (a) $[Co(NH_3)_6]\ Cl_3$ (b) $K_4[Fe(CN)_6]$.

22. Write the composition for Brass & Bronze.
23. Write the requirements for a compound to be anti aromatic.

24. Draw the structure of pyridine and furan.

25. State Kohlrausch's law.

26. What is cell constant?

27. Write a short note on current-voltage curve.

28. What is migration current?

29. What are analgesics? Give examples.

30. What is crude oil?

31. State first law of thermodynamics.

32. Define resolution.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Write the postulates of Werner's theory. **(4)**
    b) Draw the structure of chlorophyll. **(3)**

34. a) Explain the mechanism of nitration of benzene. **(4)**
    b) Explain the mesomeric effect with example. **(3)**

35. a) Write the advantages of DME. **(4)**
    b) Write the principle of potentiometric titration. **(3)**

36. a) Write the differences between analgesics, tranquilizers. **(4)**
    b) Write a note on biofuels and give examples. **(3)**

37. a) Explain the structures of Caro's acid & Marshal's acid and mention their uses. **(4)**
    b) Write a note on Bayer's strain theory. **(3)**

38. a) Write a short note on petroleum refining. **(4)**
    b) Derive the rate equation for first order reaction. **(3)**

39. a) Define phase. **(2)**
    b) Define component. **(2)**
    c) State third law of thermodynamics. **(3)**

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